Renewable Intermittency and Grid Reliability

An Endogeneous Learning Model for Vietnam

Midterm Progress Report

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Spring 2026

Presentation Outline

1. Motivation & Research Questions
2. Model Environment
3. Model Mechanics & Log-Linearization
4. Preliminary Results
5. Work in Progress

Vietnam's Energy Trilemma

Vietnam's growth model confronts three simultaneously binding constraints:

- **Energy security:** rapidly rising industrial demand
- **Sustainability:** Paris Agreement + Net Zero pledges
- **Affordability:** administratively regulated retail tariffs (EVN)

PDP8 at a Glance

	2024 actual	**2030 target**
Solar + Wind	22.8 GW	73.5 GW
Battery BESS	0.25 GW	14.0 GW
Grid invest.	\$3.2 B/yr	\$3.6 B/yr

The Core Structural Friction

Renewable generation is *non-dispatchable*: supply clears in real time with demand. As VRE penetration rises without adequate storage, the **reliability penalty**—endogenous reduction in capital utilization from grid instability—acts like a persistent negative TFP shock.

Empirical Trigger

Solar capacity doubled in 18 months (2020–21) under FiT incentives, while battery storage remained near zero. Vietnam already experiences curtailment >10% in southern provinces.

Literature Gap: What Existing Models Miss

What the literature covers:

- *E-DSGE models* (Heutel 2012; Airaudo et al. 2025): carbon taxes, "greenflation"
- *Directed Technical Change* (Acemoglu et al. 2012): R&D-race between sectors
- *Techno-economic* (Zakeri & Syri 2015; Schreiner 2020): storage costs, grid investment
- *Open economy* (Schmitt-Grohé 2003): debt-elastic interest rates

Critical blind spots:

- No model *endogenizes grid reliability* as a capital utilization constraint
- DTC literature ignores *deployment-driven learning* in storage
- Public–private investment *asymmetry* ("agility gap") is not formalized
- Open-economy E-DSGE ignores *battery import price* as a terms-of-trade shock

This capstone bridges all four gaps jointly—in a single DSGE calibrated to Vietnam.

Research Questions & Contributions

Four Research Questions

1. How does VRE intermittency propagate through capital utilization into macroeconomic output losses?
2. What is the structural timing mismatch between private battery and public grid investment—the **Agility Gap**?
3. How does endogenous learning-by-doing in storage depend on regulatory price signal transmission (χ)?
4. What are the welfare costs of intermittency and how do policy instruments rank in closing the gap?

Contribution 1

Endogenous reliability function: $u_t = f(\text{flexibility/VRE vol})$

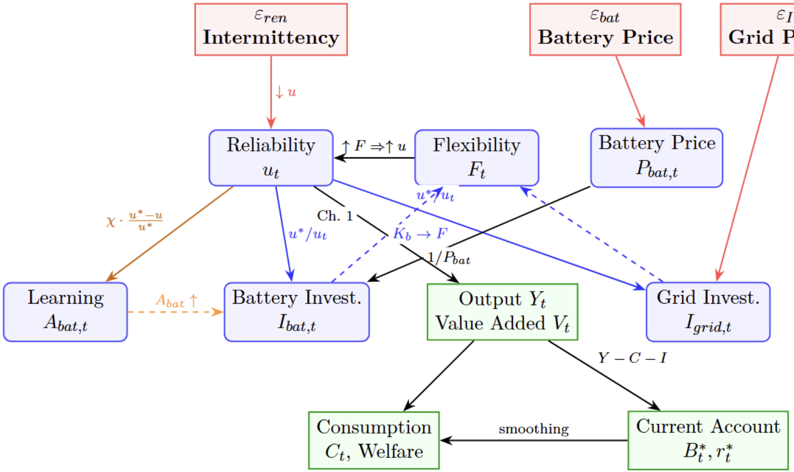
Contribution 2

Reliability-gap learning: ΔA_{bat} responds to scarcity signal χ

Contribution 3

CES agility gap: private batteries (agile) vs. public grid (sluggish)

Model Architecture: Small Open Economy DSGE



Households

Objective

- Representative household maximizes discounted utility over consumption C_t and labor L_t
- Standard log utility in consumption; disutility of labor
- Discount factor $\beta = 0.990$ (4% annualized real return)

Optimality Conditions

- **Capital Euler:** equates marginal cost of saving today to discounted return on capital tomorrow
- **Bond Euler:** links domestic consumption growth to world interest rate r_t^*
- **Labor supply:** wages equal marginal disutility of work

Budget Constraint

- Income: wages $w_t L_t$, capital rental $R_t K_{p,t}$, foreign bond returns
- Expenditure: consumption, productive investment, **battery purchases** $P_{bat,t} I_{bat,t}$, taxes, foreign bonds

Key modeling choice: Capital utilization u_t is *not* a household decision. It is an exogenous physical constraint set by grid reliability—not a margin households can optimize over.

Production & The Two Novel Equations

- Output is a CES aggregate of value-added V_t and energy $\bar{E}$ (substitutability $\sigma_E = 0.6$)
- Value-added is Cobb-Douglas with capital share $\alpha = 0.35$ — but effective capital is $u_t K_{p,t}$, so grid reliability directly reduces productive capacity

Contribution 1 — Reliability Constraint

$$u_t = 1 - \exp\left(-\psi \frac{F_t}{\phi_{int} \cdot \mathrm{Vol}_{ren,t}}\right)$$

Utilization rises as the *flexibility stock* F_t grows relative to VRE volatility. The exponential form captures nonlinear reliability costs. When $F_t \uparrow$ or volatility $\downarrow$, $u_t \to 1$.

Contribution 2 — CES Flexibility Bundle

$$F_t = \left[\mu(A_{bat,t}K_{b,t})^{\frac{\rho_f - 1}{\rho_f}} + (1-\mu)K_{g,t}^{\frac{\rho_f - 1}{\rho_f}}\right]^{\frac{\rho_f}{\rho_f - 1}}$$

Batteries and grid are *complements* ($\rho_f = 0.4 < 1$): neither can substitute for the other when the opposing asset is scarce. $A_{bat,t}$ captures endogenous learning.

Battery Investment & Endogenous Learning

Battery Investment (Private, Agile)

- Investment responds *contemporaneously* to the reliability gap $(u^* - u_t)$: the larger the shortfall, the more private capital is deployed
- Dampened by the world battery price $P_{bat,t}$ — Vietnam is a price-taker on global lithium markets
- Capital accumulates each period net of 12% annual depreciation (Li-ion lifecycle)

Contribution 3 — Endogenous Learning

- Battery productivity $A_{bat,t}$ improves when the reliability gap is large: scarcity drives learning-by-doing
- Learning speed $\eta_{bat} = 0.10$ anchored to the 18–22% Li-ion experience curve
- $\chi \in [0, 1]$: regulatory transmission parameter — how well market prices signal scarcity
- **Vietnam ($\chi = 0.3$):** regulated EVN tariffs suppress the scarcity signal, slowing battery cost reduction

Agility Gap: Private batteries respond to reliability shortfalls *immediately*; public grid investment lags 5–7 years due to permitting and construction. This asymmetry generates the mid-transition reliability valley.

Three Transmission Channels

Channel 1 — The Reliability Wedge (Output Loss)

An intermittency shock raises VRE volatility $\Rightarrow$ reliability u_t falls $\Rightarrow$ effective capital $u_t K_{p,t}$ shrinks $\Rightarrow$ output Y_t falls. Acts like a persistent, supply-side negative TFP shock, amplified by energy–capital complementarity.

Channel 2 — The Agility Gap (Asymmetric Investment)

When $u_t < u^*$: private batteries invest *immediately* (agile); public grid investment responds but faces 5–7 year construction lags (sluggish). The timing mismatch generates a predictable mid-transition reliability valley.

Channel 3 — Reliability-Gap Learning (Innovation)

A reliability shortfall triggers endogenous battery cost reduction via learning-by-doing. The speed of learning depends on χ: regulated tariffs suppress scarcity signals, slowing the innovation response and deepening the "valley of death" for storage technology.

Work in Progress

Completed ✓

- Full model derivation (16-equation equilibrium)
- Analytical steady-state characterization & solver
- Dynare implementation & Blanchard-Kahn verification
- Impulse response functions (3 shocks)
- Deterministic transition simulation (Reliability Valley)
- Log-linearization & transmission channel analysis

In Progress / Remaining

- Full calibration (structural + reliability + LBD blocks)
- **Welfare analysis:** consumption-equivalent variation $\Lambda(\chi)$
- **Counterfactual experiments:** battery subsidy ($P_{bat} = 0.8$); accelerated grid ($\phi_{grid} = 3.0$); signal suppression ($\chi = 0.3$)
- **"Perfect Storm" scenario:** joint $\varepsilon_{ren} + \varepsilon_{bat}$ simulation
- **Sensitivity analysis:** ψ, μ, β, ϕ_{grid}
- **Formal propositions:** prove $\Lambda(\chi)$ monotonicity and CES supermodularity
- **Policy section:** map $\chi \rightarrow$ concrete EVN reform agenda
- **LaTeX manuscript:** finalize bibliography & appendix proofs

Summary: Three Contributions to the Literature

1. **Endogenous Grid Reliability as a Macro Friction**

   Capital utilization $u_t = 1 - \exp(-\psi F_t/\phi_{int}\mathrm{Vol}_{ren,t})$ is determined endogenously by the stock of flexibility assets relative to VRE volatility. This transforms an engineering concept into a macroeconomic wedge in the aggregate production function—a channel absent from all extant E-DSGE models.

2. **Reliability-Gap-Induced Learning-by-Doing**

   Battery technology evolves endogenously: $\Delta A_{bat} = \eta_{bat} \cdot \chi \cdot (u^* - u)/u^*$. Unlike DTC models that require explicit R&D sectors, innovation here responds to *deployment-driven scarcity*. The regulatory parameter χ maps directly to observable policy instruments. **Policy Rec:** Vietnam must establish dynamic capacity markets (raising χ) to unlock storage cost reductions.

3. **Identification of the Structural Agility Gap**

   Private batteries (agile, price-driven) and public grid capital (sluggish, rule-based) are formally distinguished as *complementary* inputs in a CES flexibility bundle ($\rho_f = 0.4$). This generates a predictable mid-transition reliability valley—a quantitative prediction unavailable in models that treat flexibility as homogeneous.

Key References

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Thank You

Renewable Intermittency and Grid Reliability:

An Endogeneous Learning Model of Vietnam

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Keywords: Environmental DSGE · Endogenous Learning · Investment Inertia
Renewable Intermittency · Energy Storage · Vietnam PDP8

Code repository: `vietnam_dsge.mod` + `run_dynare.m` (MATLAB/Dynare 6)

Appendix: Household Equations

Lifetime Utility & Budget Constraint

$$\max \ \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\log C_t - \frac{L_t^{1+\sigma_L}}{1+\sigma_L} \right]$$

$$C_t + I_{p,t} + P_{bat,t} I_{bat,t} + B_t^* + T_t$$
$$= w_t L_t + R_t K_{p,t} + (1 + r_{t-1}^*) B_{t-1}^*$$

Optimality Conditions

Capital Euler:

$$\frac{1}{C_t} = \beta \, \mathbb{E}_t \left[\frac{\alpha Y_{t+1}/K_{p,t+1} + 1 - \delta_p}{C_{t+1}} \right]$$

Labor Supply:

$$(1-\alpha) V_t = \varphi \, C_t \, L_t^{1+\sigma_L}$$

Appendix: Production Technology

Final output (CES, $\sigma_E = 0.6$):

$$Y_t = \left[s_V \, V_t^{(\sigma_E - 1)/\sigma_E} + (1 - s_V)\bar{E}^{(\sigma_E - 1)/\sigma_E}\right]^{\sigma_E/(\sigma_E - 1)}$$

Value-added (Cobb-Douglas with endogenous utilization):

$$V_t = (u_t K_{p,t})^{\alpha} \, L_t^{1-\alpha}, \quad \alpha = 0.35$$

Investment rules:

$$\frac{I_{bat,t}}{\bar{I}_{bat}} = \left(\frac{u^*}{u_t}\right)^{\phi_{grid}} \frac{1}{P_{bat,t}} \qquad \hat{I}_{grid,t} = -\phi_{grid} \, \hat{u}_t + \varepsilon_{I,t}$$

Learning-by-doing:

$$A_{bat,t} - A_{bat,t-1} = \eta_{bat} \cdot \chi \cdot \frac{u^* - u_t}{u^*}$$

Appendix: External Sector Equations

Current account / resource constraint:

$$B_t^* = (1 + r_{t-1}^*)B_{t-1}^* + Y_t - C_t - I_{p,t} - P_{bat,t} I_{bat,t} - I_{grid,t}$$

Debt-elastic interest premium (Schmitt-Grohé 2003):

$$r_t^* = \bar{r} + \phi_b \left(e^{\bar{B}^* - B_t^*} - 1 \right), \quad \phi_b = 0.039$$

Blanchard-Kahn eigenvalues:

$\lambda_1 = 1.033$, $\lambda_2 = -2.229$, $\lambda_3 = \infty$ (3 unstable) matching C_t, Y_t, r_t^*
7 stable eigenvalues: 0, ≈ 0, 0.90, 0.97, 0.98, 0.99, 0.99 $\Rightarrow$ unique saddle-path ✓

Appendix: Full System of Log-Linearized Equations

The complete 16-equation linearized system (hat notation: $\hat{x}_t \equiv \ln X_t - \ln X_{ss}$):

1. Capital Euler: $\hat{C}_t = \mathbb{E}_t \hat{C}_{t+1} - \frac{\alpha r_{p,ss}}{1+r_{p,ss}}(\mathbb{E}_t \hat{Y}_{t+1} - \mathbb{E}_t \hat{K}_{p,t+1})$
2. Bond Euler: $\hat{C}_t = \mathbb{E}_t \hat{C}_{t+1} - \frac{r^*_{ss}}{1+r^*_{ss}} \mathbb{E}_t \hat{r}^*_{t+1}$
3. Labor: $\hat{V}_t = \hat{C}_t + (1+\sigma_L)\hat{L}_t$
4. CES output: $\hat{Y}_t = s_V^{\sigma_E} \hat{V}_t$
5. Value-added: $\hat{V}_t = \alpha(\hat{u}_t + \hat{K}_{p,t}) + (1-\alpha)\hat{L}_t$
6. Reliability: $\hat{u}_t = -\xi \hat{F}_t + \xi \hat{\mathrm{Vol}}_{ren,t}$
7. Flexibility: $\hat{F}_t = s_b(\hat{A}_{bat,t} + \hat{K}_{b,t}) + (1-s_b)\hat{K}_{g,t}$
8. Battery invest: $\hat{I}_{bat,t} = -\phi_{grid}\hat{u}_t - \hat{P}_{bat,t}$
9. Battery capital: $\hat{K}_{b,t+1} = (1-\delta_b)\hat{K}_{b,t} + \frac{I_{bat,ss}}{K_{b,ss}}\hat{I}_{bat,t}$
10. Learning: $\hat{A}_{bat,t} - \hat{A}_{bat,t-1} = -\eta_{bat} \cdot \chi \cdot \hat{u}_t$
11. Grid invest: $\hat{I}_{grid,t} = -\phi_{grid}\hat{u}_t + \varepsilon_{I,t}$
12. Grid capital: $\hat{K}_{g,t+1} = (1-\delta_g)\hat{K}_{g,t} + \frac{I_{grid,ss}}{K_{g,ss}}\hat{I}_{grid,t}$
13. Prod. capital: $\hat{K}_{p,t+1} = (1-\delta_p)\hat{K}_{p,t} + \frac{I_{p,ss}}{K_{p,ss}}\hat{I}_{p,t}$
14. Current account: [linearized B^*_t equation]
15. Interest premium: $\hat{r}^*_t = -\frac{\phi_b}{r^*_{ss}} B^*_t$
16. Battery price AR(1): $\hat{P}_{bat,t} = \rho_{bat}\hat{P}_{bat,t-1} + \varepsilon^{bat}_t$

Appendix: Welfare Cost Formula

Consumption-Equivalent Variation Λ (Lucas 1987)

Permanent % reduction in SS consumption leaving the household indifferent between the stochastic and deterministic economies:

$$\Lambda = 1 - \exp\left[\frac{1-\beta}{1-\beta^{T+1}} \sum_{t=0}^{T} \beta^t \log\left(\frac{C_t^{shock}}{C_{ss}}\right)\right]$$

$T = 40$ quarters; C_t^{shock} follows the IRF path.

Proposition: $\Lambda(\chi)$ Strictly Decreasing

$\chi \uparrow \Rightarrow$ faster $A_{bat} \Rightarrow$ higher F_t, u_t, Y_t, $C_t \Rightarrow$ lower Λ.

Proof: monotone comparative statics on linearized innovation equation.

Preliminary Result

χ	Λ	Δ from baseline
1.0 (baseline)	0.000054%	—
0.5 (partial)	0.000063%	+17%
0.0 (suppressed)	0.000073%	+35%

Interpretation Caveat

Λ magnitudes are small because linearization smooths the nonlinear reliability valley. *Relative* rankings across χ are robust. Full nonlinear welfare costs will be computed at the final defense.

Public Grid Investment & External Sector

Grid Investment (Government, Sluggish)

- *Reactive* rule: government invests more when reliability falls below target, with response strength $\phi_{grid} = 1.5$
- Financed by fixed proportional taxes ($\tau_{ss} \approx 1.43\%$); Ricardian equivalence fails
- Implementation delay shock $\varepsilon_{I,t}$ captures budget cycles, permitting, and construction lags (5–7 years)
- Grid capital depreciates at 5% annually (transmission infrastructure)

Small Open Economy (Schmitt-Grohé 2003)

- Vietnam borrows and lends at a world interest rate $\bar{r}$ subject to a debt-elastic premium
- Battery storage is *imported* at world price $P_{bat,t}$ — a direct terms-of-trade shock channel
- **"Twin drain":** an intermittency shock simultaneously reduces output *and* raises battery import demand, hitting the current account twice
- Higher foreign debt raises the sovereign borrowing rate, feeding back into domestic investment

Exogenous Shocks & Equilibrium

Three Exogenous Driving Forces

- **Intermittency shock ε_{ren}:** AR(1) with persistence $\rho_{ren} = 0.85$ (monsoon seasonality); volatility $\sigma_{ren} = 0.12$ (CV of monthly VN solar output)
- **Battery price shock ε_{bat}:** AR(1) with persistence $\rho_{bat} = 0.90$; volatility $\sigma_{bat} = 0.08$ (lithium carbonate cycle, BNEF 2023)
- **Grid policy shock ε_{I}:** implementation delays from budget cycles and permitting; mutually independent of other shocks

Competitive Equilibrium: 16 Equations, 16 Unknowns

Block	Equations
Households	Capital Euler, Bond Euler, Labor supply
Production	CES output, Value-added
Energy/Grid	Reliability u_t, Flexibility F_t
Battery	I_{bat} rule, K_b accumulation, LBD A_{bat}
Grid	I_{grid} rule, K_g accumulation
Capital	K_p accumulation
External	Current account, Interest premium
Shocks	Battery price AR(1)

Blanchard-Kahn verified: 3 eigenvalues outside the unit circle match the 3 forward-looking variables (C_t, Y_t, r_t^*). Unique saddle-path equilibrium confirmed.

Calibration Strategy: Three-Stage Approach

1. **Standard structural parameters** matched to long-run Vietnamese macroeconomic aggregates (GSO 2024, ADB, World Bank)
2. **Reliability & intermittency parameters** micro-founded on engineering data: NLDC facility-level generation, IEA Vietnam Energy Outlook 2023, EVN annual reliability reports
3. **Learning-by-doing parameters** anchored to empirical experience curve literature for Li-ion batteries (Kittner et al. 2017; Way et al. 2022)

Calibration Target: Flexibility Weight $\mu = 0.16$

From PDP8 investment allocations:

$K_{b,ss} = \frac{2.4/10}{0.12} \approx 2.0$ B USD (BESS)

$K_{g,ss} = \frac{14.9/10}{0.05} \approx 29.8$ B USD (grid)

With $\rho_f = 0.4$: $\mu \approx 0.16$. Reflects the *small but growing* share of private storage in Vietnam's current infrastructure.

Complementarity: $\rho_f = 0.40$

Transmission moves energy across *space*; storage moves it across *time*. When solar output is zero (night), infinite transmission capacity is useless. Hart (2019) and Hirth (2013) document strong complementarity empirically—$\rho_f = 0.4$ enforces this binding constraint.

Calibrated Parameters (I): Preferences, Production & Energy

Category	Parameter	Value	Source / Rationale
Preferences & Production	Discount factor β	0.990	Annualized real return $\approx$ 4% (VGB)
	Capital share α	0.350	Labor income share = 65% (GSO 2024)
	Depreciation (prod.) δ_p	0.025	10% annually (standard)
	Labor elasticity σ_L	1.000	Frisch elasticity (NK literature)
	Energy-VA elasticity σ_E	0.600	Heutel (2012); energy–capital complementarity
Energy & Reliability	Reliability sensitivity ψ	2.000	Moment-matched: $u_{ss} = 97\%$ (EVN reports)
	Intermitt. persistence ρ_{ren}	0.850	Monsoon AR(1) (IEA 2023)
	Intermitt. volatility σ_{ren}	0.120	CV of monthly VN solar output (GSO 2024)
	Battery weight μ	0.160	PDP8 BESS/grid capital stock ratio
	Flexibility elasticity ρ_f	0.400	Complementarity (Hart 2019; Hirth 2013)
	Depreciation (battery) δ_b	0.030	12% annually (Li-ion lifecycle)
	Depreciation (grid) δ_g	0.0125	5% annually (transmission infrastructure)

Calibrated Parameters (II): Innovation, Shocks & External Sector

Category	Parameter	Value	Source / Rationale
Learning-by-Doing	Learning elasticity η_{bat}	0.100	18–22% Li-ion exp. curve (Kittner 2017; Way 2022)
	Signal transmission χ	1.0	Baseline: fully liberalized market
		0.3	Counterfactual: regulated tariff regime
Battery Price Shock	Persistence ρ_{bat}	0.900	AR(1) of lithium carbonate (BNEF 2023)
	Volatility σ_{bat}	0.080	Li-ion pack price survey (BNEF 2023)
Fiscal Policy	Grid invest. response ϕ_{grid}	1.500	Reactive lag structure (World Bank 2022)
External Sector	World interest rate $\bar{r}$	0.0101	$= 1/\beta - 1$ (quarterly)
	Debt premium ϕ_b	0.039	Schmitt-Grohé & Uribe (2003)
	SS net foreign assets $\bar{B}^*$	0	Balanced trade normalization

Calibration Validation: Model Fit to Vietnamese Data

Moment	Data	Model	Source
Investment rate (I/Y)	27–29%	27.2%	GSO 2024; targeted
Consumption share (C/Y)	64–66%	72.8%†	GSO 2024; model SS
Grid reliability (utilization)	95–98%	97.0%	EVN reports; targeted
Annualized real interest rate	≈ 4%	4.04%	VGB average

† *Note:* Model abstracts from government consumption (≈ 6–7% of GDP). Adjusted private consumption share ≈ 66%—consistent with data.

Reliability Parameter Validation

The steady-state utilization shortfall maps to Energy Not Served (ENS):

$$(1 - u_t) \approx \mathrm{ENS}_t / \mathrm{Demand}_t$$

$u_{ss} = 97\%$ implies 3% ENS—within EVN's reported 3–5% range.

Sensitivity Check

Results are robust for $\psi \in [1.5, 3.0]$ and $\mu \in [0.20, 0.35]$. The *qualitative* Agility Gap mechanism holds across the full parameter space tested.

Steady State & Solution Method

Steady-State Values

Variable	Value	Interpretation
Y_{ss}	0.868	Normalized
C_{ss}	0.631	72.8% of output
$K_{p,ss}$	8.651	Productive capital
$K_{b,ss}$	0.190	Battery capital
$K_{g,ss}$	1.140	Grid capital
u_{ss}	0.970	Grid reliability
r^*_{ss}	0.0101	World rate

Solution Method

- Model solved by **first-order perturbation** around the deterministic steady state using *Dynare 6*
- System expressed in log-deviations from steady state ($\hat{x}_t \equiv \ln X_t - \ln X_{ss}$)
- Full log-linearized system: 16 equations, 16 unknowns (see Appendix)

Blanchard-Kahn Verified

3 eigenvalues outside the unit circle match the 3 forward-looking variables (C_t, Y_t, r^*_t). Unique saddle-path equilibrium confirmed. ✓

The exponential nonlinearity of u_t is absorbed into linearization coefficient ξ. Nonlinear transition dynamics are analyzed

Baseline Results: Steady State & FEVD Overview

Forecast Error Variance Decomposition (40Q)

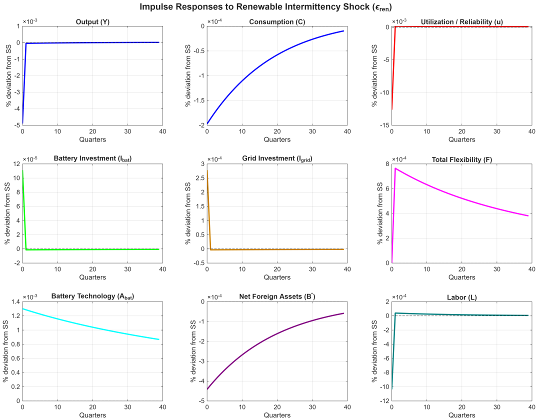
Variable	ε_{ren}	ε_{bat}	ε_{I}
Output (Y)	**79%**	16%	5%
Reliability (u)	**85%**	12%	3%
Consumption (C)	11%	**80%**	9%
Battery invest.	1%	**100%**	<1%
Grid invest.	14%	1%	**85%**

Three Structural Findings from FEVD

1. **Intermittency dominates** output (79%) and reliability (85%)—validates the model's core reliability channel
2. **Battery investment** is 100% driven by global price shocks, not domestic policy—Vietnam is a price-taker
3. **Consumption** is dominated by battery price shocks (80%), not intermittency—household smooths domestic shocks via borrowing but is exposed to import price volatility

Note: The B^* shares exceed 100% due to negative covariance: intermittency drains the current account (output falls) *and* battery price shocks raise import costs simultaneously.

Impulse Response: Renewable Intermittency Shock



One standard deviation shock to renewable intermittency ($\sigma_{ren} = 0.12$). All variables: % deviation from steady state over 40 quarters.

Impulse Response: Three-Stage Transmission Narrative

Stage 1 — Impact (Q0)

$u_t \downarrow -0.013\%$; $Y_t \downarrow -0.005\%$ $\Rightarrow$ Reliability penalty activates immediately via the CES output wedge.

Stage 2 — Investment Response (Q1–Q10)

$I_{bat} \uparrow +0.0001\%$ (*agile, contemporaneous*) vs. $I_{grid} \uparrow +0.0003\%$ (*sluggish, lagged*)

Agility gap is visible: private batteries respond faster than public grid. Battery import demand rises $\Rightarrow$ current account deteriorates ("twin drain").

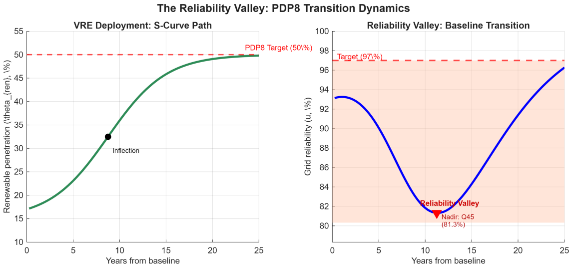
Stage 3 — Learning & Recovery (Q10+)

$A_{bat} \uparrow +0.11\%$ by Q10; endogenous LBD activated.

Gradual reliability recovery as flexibility stock accumulates. Full mean-reversion by Q30–Q35 under baseline $\chi = 1$.

Key takeaway: The intermittency shock is persistent ($\rho_{ren} = 0.85$) but the endogenous learning channel ($\chi > 0$) shortens the recovery path. Regulated markets ($\chi = 0.3$) slow recovery significantly—the counterfactual welfare cost is shown in the Appendix.

The Reliability Valley: PDP8 Transition Path (Preliminary)



Deterministic simulation: VRE penetration 15%→50% over 100 quarters (S-curve). Reliability falls as VRE outruns flexibility accumulation.

Valley Statistics

Nadir at:	Quarter 45
Min. reliability:	81.3%
Below target:	15.7 pp
Implied ENS:	≈ 1,640 hrs/yr

Three Causes

1. Front-loaded VRE (FiT boom)
2. Lagged grid (5–7 yr permitting)
3. Battery supply constraints

Full welfare analysis and policy counterfactuals in progress—to be presented at final defense.